

Assignment 01

Cloud Computing

Submitted To: Engr.Waqas Saleem

Submitted By: Faiqa Shahzad

Date: October 10, 2026

Day: Saturday

Semester: Fifth

Registration No: 2024-BSE-26

Department of Software Engineering

(2024-2028)

Task 1:

Install Gitea and Push a Repository from the Ubuntu Server

Screenshots:

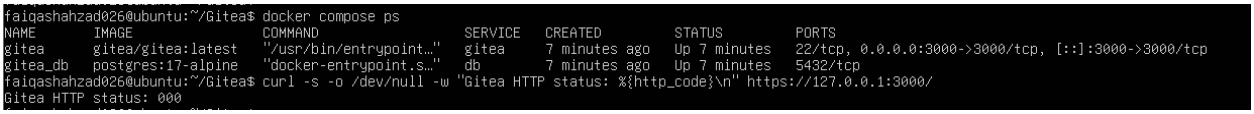


Figure 1.1 task1_gitea_running.png

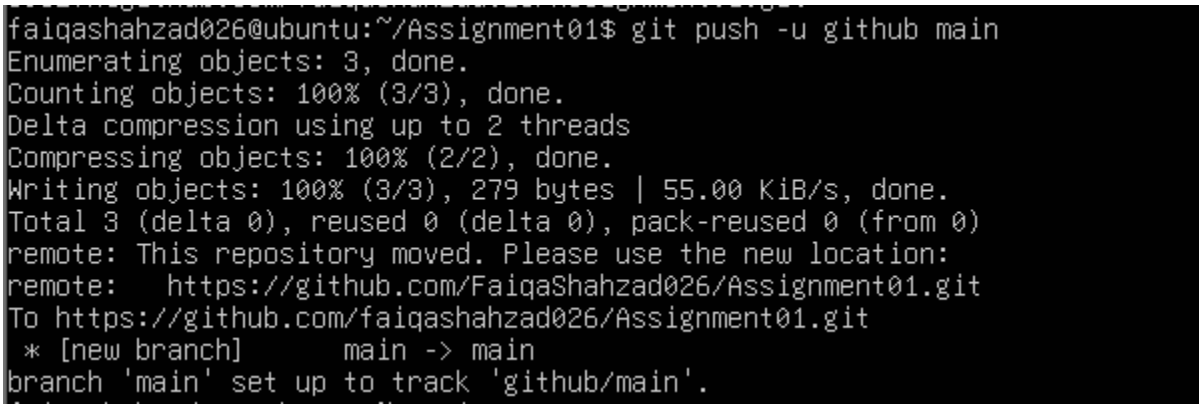


Figure 1.2 task1_gitea_push.png

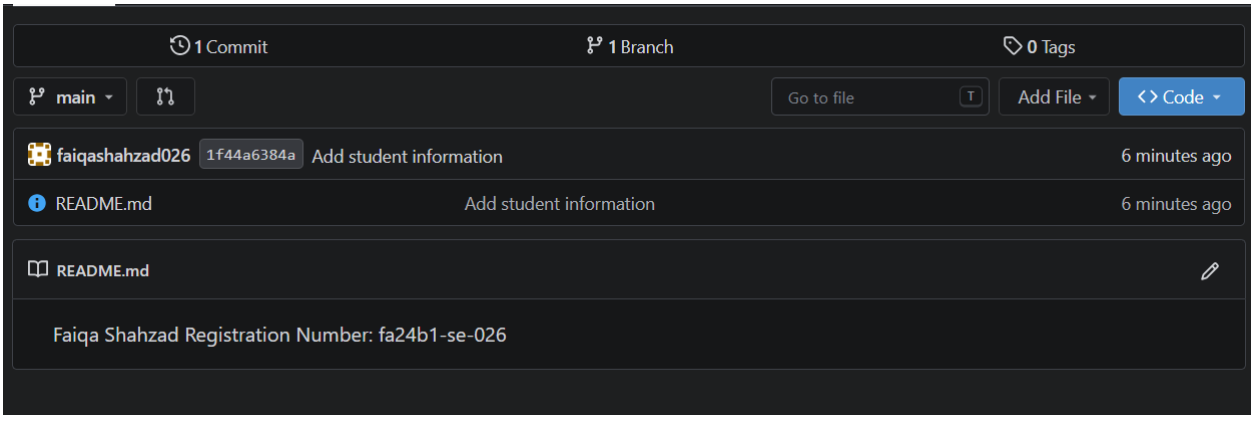


Figure 1.3 task1_gitea_repository.png

Task 2:

Push the Same Repository to GitHub

Screenshots:

```
faiqashahzad026@ubuntu:~$ cd ~/Assignment01
faiqashahzad026@ubuntu:~/Assignment01$ git remote add github https://github.com/faiqashahzad026/Assignment01.git
faiqashahzad026@ubuntu:~/Assignment01$ git remote -v
gitea   http://faiqashahzad026@127.0.0.1:3000/faiqashahzad026/Assignment01.git (fetch)
gitea   http://faiqashahzad026@127.0.0.1:3000/faiqashahzad026/Assignment01.git (push)
github  https://github.com/faiqashahzad026/Assignment01.git (fetch)
github  https://github.com/faiqashahzad026/Assignment01.git (push)
```

Figure 2.1 task2_remotes.png

```
faiqashahzad026@ubuntu:~/Assignment01$ git push -u github main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 279 bytes | 55.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   https://github.com/FaiqaShahzad026/Assignment01.git
To https://github.com/faiqashahzad026/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
```

Figure 2.2 task2_github_push.png

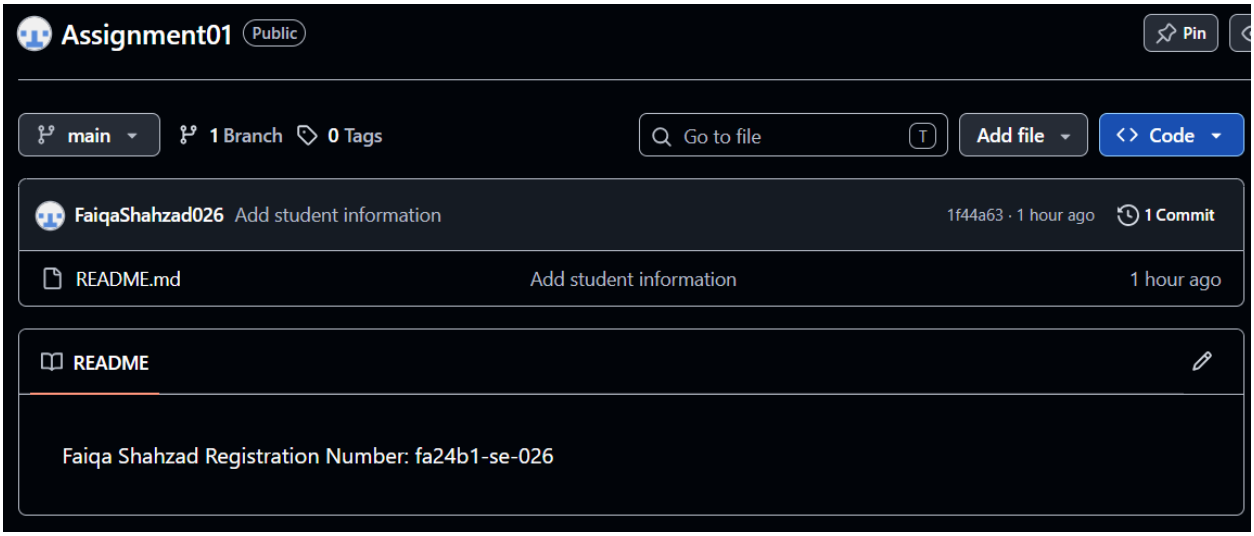


Figure 2.3 task2_github_repository.png

Task 3:

Track Three Large Files with Git LFS

Screenshots:

```
faiqashahzad026@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
faiqashahzad026@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
faiqashahzad026@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
faiqashahzad026@ubuntu:~/Assignment01$ cat .gitattributesS
cat: .gitattributesS: No such file or directory
faiqashahzad026@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
```

Figure 3.1 task3_lfs_setup.png

```
faiqashahzad026@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
faiqashahzad026@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
faiqashahzad026@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
faiqashahzad026@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
faiqashahzad026@ubuntu:~/Assignment01$ git lfs ls-files --size

f1753a034b * Assignment01/large-file-1.bin (106 MB)
2338a970c0 * Assignment01/large-file-2.bin (107 MB)
cb91d5908e * Assignment01/large-file-3.bin (108 MB)
```

Figure 3.2 task3_lfs_files.png

```
faiqashahzad026@ubuntu:~/Assignment01$ git commit -m "Add three files using Git LFS"
[main 7067305] Add three files using Git LFS
 4 files changed, 10 insertions(+)
 create mode 100644 Assignment01/.gitattributes
 create mode 100644 Assignment01/large-file-1.bin
 create mode 100644 Assignment01/large-file-2.bin
 create mode 100644 Assignment01/large-file-3.bin
faiqashahzad026@ubuntu:~/Assignment01$ git push github main
Uploading LFS objects:   0% (0/3), 178 MB | 2.1 MB/s
Uploading LFS objects: 100% (3/3), 321 MB | 2.0 MB/s, done.
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 851 bytes | 212.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   https://github.com/FaiqaShahzad026/Assignment01.git
To https://github.com/faiqashahzad026/Assignment01.git
   1f44a63..7067305  main -> main
```

Figure 3.3 task3_lfs_push.png

Task 4:

Create a Portfolio or CV with GitHub Pages

Screenshots:

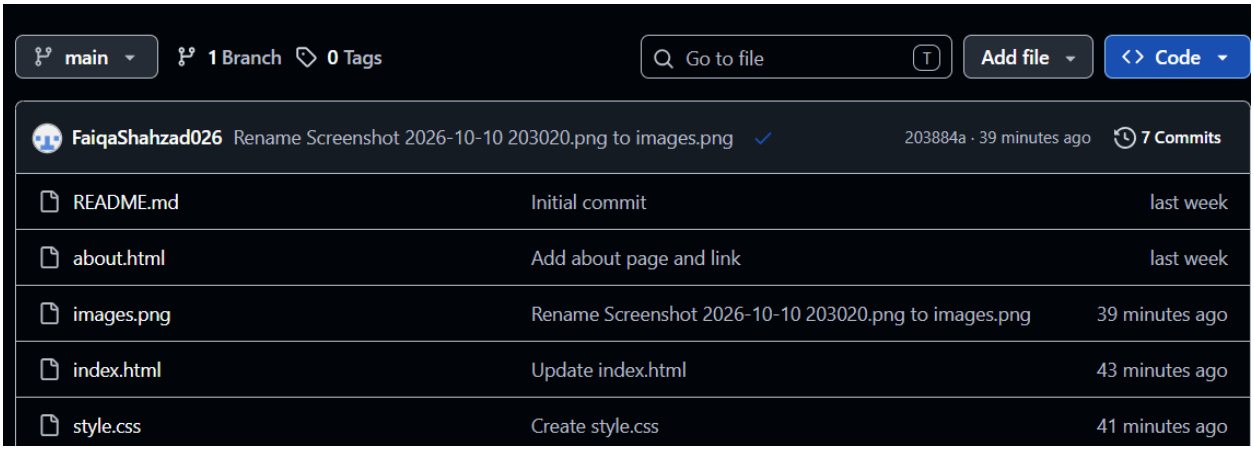


Figure 4.1 task4_pages_repository.png

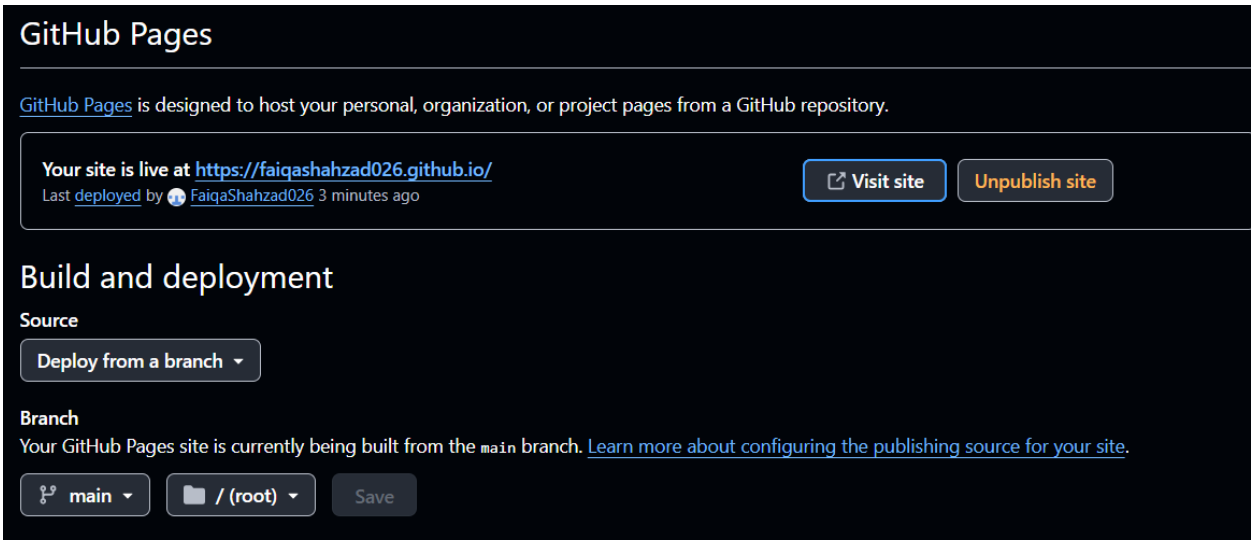


Figure 4.2 task4_pages_deployment.png

FAIQA SHAHZAD

Software Engineering & Cloud Computing Student

GitHub: github.com/faiqashahzad026 | **Location:** Islamabad, Pakistan | About Page

PROFESSIONAL SUMMARY

Motivated Software Engineering student with strong academic foundation in cloud computing, relational database management, object-oriented system design, and computer networking. Proven experience in developing desktop applications, creating comprehensive UML architectural models, and deploying automated cloud version control pipelines.

TECHNICAL SKILLS

Languages & Frameworks:	C#, SQL, HTML5, CSS3
Cloud & DevOps:	GitHub Pages, Git LFS, Gitea Mirroring, Git Version Control
Databases & Networking:	Microsoft SQL Server, ER/EER Modeling, Cisco Packet Tracer (OSPF, VLANs)
Software Design:	UML Modeling (Use Case, Class, Sequence, Component, Deployment)

KEY PROJECTS

Cloud Infrastructure & Automated Static Portfolio Deployment *2026*

- Configured and deployed a live portfolio application via **GitHub Pages** with custom routing and domain structure.
- Integrated **Git LFS (Large File Storage)** to handle heavy media binary tracking seamlessly across repositories.
- Established remote mirror synchronization between local Git environments, GitHub, and self-hosted Gitea instances.

Desktop Shopping System (C# & Windows Forms) *2026*

- Designed an object-oriented C# desktop GUI using Windows Forms and DataGridView components.
- Integrated Microsoft SQL Server back-end for dynamic product inventory, user transactions, and order management.

Multi-Router Ring Network Topology Setup *2026*

- Simulated multi-router network architecture with **OSPF dynamic routing** and switch port security using Cisco Packet Tracer.
- Configured VLAN segmentation, IP addressing, and serial inter-router interfaces via Cisco IOS CLI.

EDUCATION & RELEVANT COURSEWORK

Bachelor of Science in Software Engineering *In Progress*

Key Coursework:

- **Software Engineering & UML Modeling:** Complete system design lifecycle with behavioral and structural UML viewpoints.
- **Database Systems:** Database normalization (1NF–3NF), ER diagramming, complex SQL joins, and script optimization.
- **Computer Networks:** Routing protocols, subnetting, TCP/IP, and switch security configurations.

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Figure 4.3 task4_portfolio_live.png